JAVA FUNDAMENTALS – DETAILED STUDENT GUIDE

---

1. Key Features of Java

Definition:
Java is a high-level, object-oriented, platform-independent programming language developed by Sun Microsystems.

Important Features

- **Platform Independent**
  - Java code is compiled into bytecode, not machine code.
  - Bytecode runs on JVM, which exists for every OS.
  - Same program runs on Windows, Linux, macOS.
- **Object-Oriented**
  - **Java follows OOP concepts:**
    - Encapsulation
    - Inheritance
    - Polymorphism
    - Abstraction
- **Automatic Memory Management**
  - Java uses Garbage Collector.
  - Programmer does not free memory manually.
- **Robust & Secure**

    - No pointer arithmetic.
    - Strong exception handling.
    - JVM verifies bytecode.
- **Multithreaded**
    - Multiple tasks can run simultaneously.

Exam Line:

Java is platform independent due to JVM and supports OOP with automatic memory management.

---

2. Primitive vs Reference Data Types

Primitive Data Types

Definition:
Primitive data types store actual values directly in memory.

Examples:

- int
- float
- char
- boolean

Characteristics:

- Fixed size
- Faster
- No methods
- Stored in stack (mostly)

Reference Data Types

Definition:
Reference data types store the address of an object, not the actual data.

Examples:

- String
- Arrays
- Objects

Characteristics:

- Store memory address
- Can call methods
- Stored in heap

Key Difference Line:

Primitive stores value, reference stores address.

---

3. Method Overloading

Definition:
Method overloading allows multiple methods with the same name but different parameters in the same class.

Rules:

- Different number of parameters
- Different data types
- Different order of parameters

❌ **Return type alone cannot overload.**

Why it is used

- Improves readability
- Supports compile-time polymorphism

---

4. Multithreading Using Runnable

Definition:
Multithreading is the ability to execute multiple threads simultaneously.

Runnable Interface

- Contains only one method:
- public void run();
- Represents a task, not a thread.
- Thread executes the task.

Execution Flow

1. Runnable object created
2. Passed to Thread
3. start() called
4. JVM scheduler runs thread

Exam Line:

Runnable is preferred over Thread because Java supports single inheritance.

---

5. Applet vs Standalone Application

Applet

Definition:
An applet is a Java program that runs inside a browser or appletviewer.

Characteristics:

- No main() method
- Uses lifecycle methods (init, start)
- Restricted environment (sandbox)
- Obsolete

Standalone Application

- Runs independently on OS
- Uses main() method
- Full system access

Difference Line:

Applet runs in browser, standalone runs on OS.

---

6. Event Listener in Java

Definition:
An event listener is an object that responds to user actions like clicks or key presses.

Event Handling Flow

1. User action
2. Event generated
3. Event object created
4. Listener method invoked

Used in:

- AWT
- Swing
- JavaFX

Exam Line:

Event handling follows the delegation event model.

---

7. JDBC (Java Database Connectivity)

Definition:
JDBC is an API that enables Java programs to interact with databases.

JDBC Components

- Driver
- Connection
- Statement / PreparedStatement
- ResultSet

Flow

Java → JDBC API → Driver → Database

Important Concept:
PreparedStatement prevents SQL injection.

---

8. == Operator vs equals() Method

== Operator

- Compares memory addresses for objects
- Compares values for primitives

equals() Method

- Compares content
- Must be overridden for custom classes

Golden Rule:

Use == for reference, equals() for data.

---

9. Types of Variables in Java

Local Variable

- Declared inside method
- No default value
- Stack memory

Instance Variable

- Declared in class
- One copy per object
- Heap memory

Static Variable

- Shared among all objects
- One copy per class

Parameter Variable

- Method arguments
- Treated as local variables

---

10. Byte Streams vs Character Streams

Byte Streams

- Handle binary data
- Classes: InputStream, OutputStream
- Used for images, audio

Character Streams

- Handle text data
- Classes: Reader, Writer
- Encoding aware

Exam Rule:

Binary → Byte stream
Text → Character stream

---

11. Difference Between C++ and Java

Object-Oriented

- C++: Partial OOP
- Java: Mostly pure OOP

Memory Management

- C++: Manual (new/delete)
- Java: Automatic (GC)

Platform

- C++: Platform dependent
- Java: Platform independent

---

12. String vs StringBuilder

String

- Immutable
- Slow for changes
- Thread-safe

StringBuilder

- Mutable
- Fast
- Not thread-safe

Exam Line:

String is immutable, StringBuilder is mutable.

---

13. Layout Manager in AWT

Definition:
Layout manager controls position and size of components in a container.

Purpose:

- Automatic arrangement
- Handles resizing
- Platform independence

Examples:

- FlowLayout
- BorderLayout
- GridLayout